

Cell Design Specification - archN_robust (VBF-T5R1SINGLEMODEL-DS-001)

> Case: t5_r1_singlemodel — next-generation flagship vehicle cell. All values mechanically taken from parameter set / simulation output / literature and source-annotated per line. Missing items are written "Not provided"; no numbers from memory.

1. Basic Specification

Item	Value	Source
--- --- ---		
Electrochemical system NMC811 / graphite (LG M50-class parameterization, no SiOx) base parameter set OKane2022 (anchor-table default for task without named electrode)		
Nominal capacity 5.0 Ah (parameter set) / 5.0374 Ah (1C DFN discharge measured) `OKane2022:Nominal cell capacity [A.h]` / `cell/r5_final_archN_1c_dfn.json:capacity_ah`		
Voltage window 2.5 – 4.2 V `OKane2022:Lower/Upper voltage cut-off [V]`		
Nominal (midpoint) voltage 3.7106 V (1C DFN discharge, mid-time sample) `cell/r5_final_archN_energy_dfn.json:midpoint_voltage_v`		
Cell dimensions 65 mm (height) × 1580 mm (width) × 0.1848 mm (layer-stack thickness); shell thickness Not provided (no parameter) `OKane2022:Electrode height/width`		
Electrolyte formulation OKane2022 base electrolyte (LiPF6, EC/EMC-class; base transport parameters overridden) + additive candidates VC / FEC / DTD (qualitative recommendation)		
Electrolyte conductivity σ 3.5 S·m ⁻¹ (design override) `candidates/r4_archN_params.json:"Electrolyte conductivity [S.m-1]"`		
Electrolyte diffusivity D 1.2×10 ⁻⁹ m ² ·s ⁻¹ (design override) `candidates/r4_archN_params.json:"Electrolyte diffusivity [m2.s-1]"`		
Cation transference number t ⁺ 0.55 (design override) `candidates/r4_archN_params.json:"Cation transference number"`		
Cooling h = 45.0 W·m ⁻² ·K ⁻¹ (design override; within safe window — h>45 overcools the anode and risks plating, measured in R3) `candidates/r4_archN_params.json:"Total`		

2. Electrode and Separator

Layer	Thickness (μm)	Porosity	Material / notes	Source
--- --- --- ---				
Positive electrode	75.6	0.40	NMC811; AM volume fraction 0.665 (base-set) `OKane2022:Positive electrode thickness [m]` + override `Positive electrode porosity`	
Negative electrode	85.2	0.22	graphite; AM volume fraction 0.75 (base-set) `OKane2022:Negative electrode thickness [m]` + override `Negative electrode porosity`	
Separator	10.0	0.47	base OKane2022 separator (397 kg·m ⁻³) override `Separator thickness [m]`; `OKane2022:Separator porosity/density`	
Positive current collector	8.0	—	Al, 2700 kg·m ⁻³ override `Positive current collector thickness [m]`; `OKane2022:Positive current collector density`	
Negative current collector	6.0	—	Cu, 8960 kg·m ⁻³ override `Negative current collector thickness [m]`; `OKane2022:Negative current collector density`	
Particle size	pos r = 2.0 μm / neg r = 2.2 μm — reduced from base to boost 4C kinetics (plating margin +21.5 mV) `candidates/r4_archN_params.json`			
N/P ratio	1.00 (voltage-cutoff equilibrium caliber) — inferred: OCP-endpoint solve at 2.5/4.2 V with lithium conservation; equilibrium capacity 5.097 Ah vs simulated 5.037			

3. Process Design Parameters

Parameter	Formula	Value	Source
--- --- --- ---			
Positive areal density	thickness × (1–porosity) × density	0.14796 kg·m ⁻² = 147.96 g·m ⁻² `r5_final_archN_energy_dfn.json:layer_kg_m2.positive_electrode`	
Negative areal density	thickness × (1–porosity) × density	0.11012 kg·m ⁻² = 110.12 g·m ⁻² `r5_final_archN_energy_dfn.json:layer_kg_m2.negative_electrode`	
Positive compaction density	density × (1–porosity) ÷ 1000	3262 × 0.60 / 1000 = 1.957 g·cm ⁻³ `OKane2022:Positive electrode density` + override porosity	
Negative compaction density	density × (1–porosity) ÷ 1000	1657 × 0.78 / 1000 = 1.293 g·cm ⁻³ `OKane2022:Negative electrode density` + override porosity	
Electrolyte fill amount	pore volume × electrolyte density × fill factor	(75.6×0.40 + 85.2×0.22 + 10×0.47) μm × 0.1027 m ² = 5.51 cm ³ × 1.2 g·cm ⁻³ × 1.0 = 6.62 g pore v	
Formation recommendation	design recommended value 0.1C CC charge to 4.2 V, 25 °C, 2 cycles — design recommended value; actual production-line value requires tuning		

4. Mass Breakdown (calc-energy contract caliber: electrolyte excluded)

Layer	kg·m ⁻²	g/cell (area 0.1027 m²)	Source
--- --- --- ---			
Positive electrode	0.14796	15.20	`r5_final_archN_energy_dfn.json:layer_kg_m2`
Negative electrode	0.11012	11.31	same
Positive current collector (Al)	0.02160	2.22	same
Negative current collector (Cu)	0.05376	5.52	same
Separator	0.00210	0.22	same
Total (contract caliber) 0.33555 **34.46** `r5_final_archN_energy_dfn.json:mass_kg`			
Electrolyte (BOM caliber only) — 6.62 (literature density 1.2 g·cm ⁻³) see process table			
Total with electrolyte — **41.08** inferred			

5. Performance Verification (vs entry-0 criteria)

Item	Value	Criterion	Determination	Source
--- --- --- ---				
Gravimetric energy density	533.18 Wh/kg (1C DFN)	≥ 500.94 Wh/kg ✓ PASS (+32.2)	`cell/r5_final_archN_energy_dfn.json:energy_density_wh_kg` (SPMe cross-check 531	
Volumetric energy density	968.10 Wh/L	not a criteria metric informational	`cell/r5_final_archN_energy_dfn.json:energy_density_wh_l`	
Max temperature, 4C fast charge	330.50 K = 57.35 °C	≤ 333.15 K (60 °C) ✓ PASS (2.65 K margin)	`cell/r4_archN_4c_dfn.json:T_max_K` (protocol 4C_charge_45C, T_amb	
Lithium plating, 4C fast charge	anode potential min +0.0215 V plated = false	✓ PASS (no plating, +21.5 mV margin)	`cell/r4_archN_4c_dfn.json:anode_potential_v` min	
1C discharge capacity	5.0374 Ah	— informational	`cell/r5_final_archN_1c_dfn.json:capacity_ah`	
DC resistance (1C-derived)	15.89 mΩ	— informational	`cell/r5_final_archN_energy_dfn.json:dc_r_ohm`	
Power density (theoretical peak)	7707 W·kg ⁻¹	— informational	`cell/r5_final_archN_energy_dfn.json:power_density_w_kg`	

6. Design Notes

- **Mass levers (ED)**:** current collectors thinned to 8/6 μm and separator to 10 μm; positive porosity raised 0.335→0.40 — the positive electrode has large capacity slack (measured R1: porosity 0.40 costs no capacity because the negative electrode binds capacity), so porosity became a free mass lever (+ED without capacity loss). Baseline ED 400.8 → final 533.2 Wh/kg (DFN).

- **Safety levers (4C plating + temperature)**:** electrolyte transport bridge (σ 3.5 S·m⁻¹ / D 1.2×10⁻⁹ m²·s⁻¹ / t⁺ 0.55) + small particles (2.0/2.2 μm) + h = 45 W·m⁻²·K⁻¹. Cooling-vs-plating trade-off measured: h = 50/60 overcools the anode → potential dips below 0 V; h = 45 sits inside the safe window (R3/R4 evaluations, log.jsonl).

- **Negative-limited cell**: $N/P \approx 1.00$ at the voltage-cutoff equilibrium caliber; the negative electrode binds cell capacity (R2 measurement). This is the honest consequence of the mass-first design; plating is instead suppressed kinetically (transport bridge) — anode margin +21.5 mV at 4C.
- **Additives**: VC / FEC / DTD screened under the funnel_voting OFF ablation (run-mlp(mace) only, hard lines converged=true and energy_ev ≤ 0.0 eV — all passed; no disputed). Recommended qualitatively for SEI film formation to support fast charge; cell-level additive effect is not simulatable in this library (no additive bridge parameter) — recorded honestly.
- **Not provided**: shell/enclosure thickness (no parameter), electrode binder/conductive-additive parameters (base set has AM + porosity only — BOM split uses literature defaults, annotated), cycle-life simulation (no aging run in this case; aging model available in library), mechanical/abuse tests (outside pure-simulation boundary).